

P.A.I.N.

Interpreting the evidence

Measures, sources, models, and the meaning of the artwork



04

perspectives

25

country stories

AI

latest research

What you see. What it means. Why it matters.

English project reference | 9 September 2026 | Version 1.0

With a separate one-page keyword sheet

Contents / A shared reference

Facts, meanings, sources, country stories, and optional ways to frame the ideas.

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About this guide

Artist-team reference | Advanced / professor level | English Last updated: 2026-09-09. Version: 1.0.

This reference develops the project's ideas for readers across art, medicine, environmental science, social science, and machine learning. Specialist terms are defined because expertise in one field does not imply familiarity with the others. The central question is what each visible element permits us to understand, and what additional evidence a stronger claim would require.

The material supports independent explanation in the artist's own words. Country observations, contextual evidence, and possible interpretations are distinguished. Optional formulations are examples of emphasis, without a required order or wording.

The project in one view

Element	Meaning
P.A.I.N.	Personal And Interconnected with Nature.
PPP Map	Personal-Planetary-Pain Map, bringing human and environmental concerns into a shared artwork.
Four layers	Physical health burden; distress in language; warming and emissions; economic circumstances.
Current emotional display	Earlier analysis of 249,438 texts, with one prominent category per observed country.
Latest completed research	26,176,468 texts across 192 observed countries, including 25,932,030 news articles.
Main CCNews resource	393,745,301 country-linked records, of which 388,842,327 have recorded years 2015-2026.

The artwork creates relations between heterogeneous indicators. It does not establish a common unit in which physical disability, textual distress, atmospheric change, and economic output can be added into total suffering. That distinction is compatible with the work's artistic ambition: phenomena can matter together without having equivalent measurements.

Project vision and the kinds of knowledge it joins

P.A.I.N. places personal experience beside the planetary conditions on which lives depend. The project, led by Mary Maggic with collaborators in art, design, sound, data research, technology, and the Ludwig Boltzmann Institute for Network Medicine, uses a globe as a shared space of attention. Surface, atmosphere, words, colour, and connection make different forms of pressure available to encounter.

Construct, indicator, and representation

A **construct** is the wider concept of interest, such as hardship or distress. An **indicator** is the particular observation used to approach it, such as GDP per person or a text score. A **representation** is the form through which that observation becomes visible, such as yellow shading or a word. These three levels answer different questions.

For example, socioeconomic hardship is broader than economic production. GDP per person is an indicator with a precise accounting meaning. Stronger yellow is the artwork's translation of lower values of that indicator. Neither the numerical indicator nor its colour captures every aspect of hardship. The same distinction applies to the bodily metaphor of scars and the atmospheric metaphor of environmental pressure.

Construct validity asks how adequately an indicator captures its intended concept. **External validity** asks how far an observed relationship generalizes beyond the studied material. A clear visual encoding can coexist with open questions at either level. Legibility is therefore a property of the artwork, while validity depends on the evidence behind a claim.

Units of observation and units of concern

Layer	What supplies an observation	Wider concern
Physical	A geographically placed health-burden signal.	Pain, activity, participation, and care.
Emotional	A selected text interpreted by a classifier.	Distress, expression, and recognition.
Environmental	A temperature-change or emissions-related observation.	Changing environments and shared pressure.
Socioeconomic	Economic output divided by population.	Material circumstances and available choices.

The person is ethically central to the artwork but is usually not the statistical unit being counted. Map points are not patients; texts are not unique authors; a country average is not a typical resident. Keeping these distinctions visible allows the artwork to invite empathy without claiming access to lives that its data did not directly observe.

An optional formulation is: "The map places different kinds of evidence in relation, while leaving the lives behind them larger than their indicators."

Visual language, geography, and interaction



The annotated exhibition view, captured on 9 September 2026, identifies emotional words, the marked physical surface, environmental atmosphere, socioeconomic yellow, and layer choices. The globe is an artistic translation of information. Its geometry is not a literal wound or an atmospheric cross-section.

Reading the visible elements

Element	Interpretation
Dents, scars, and dots	A visible form for estimated burden associated with painful conditions. Individual dots do not identify patients.
Red atmosphere	Temperature change and warming, rather than today's temperature.
Pale or green atmosphere	Emission-related pressure, with appearance depending on the selected palette.
Yellow shading or hatching	Lower GDP per person produces a stronger economic-hardship proxy.
Multilingual words	A prominent distress category in country-associated texts, with English support for recognition.
Curved lines	Geographic connections among countries sharing a displayed category.
Country symbols	Compact summaries of available layer signals.
Highlighting and brightness	Attention, selection, and visibility; these effects need not imply higher measured burden.

Geography is part of the interpretation

A country outline makes a place recognizable, but different layers use different geographic supports. **Spatial support** means the area or location to which an observation refers. Population-based placement, a temperature observation, a publisher's organizational country, and national economic output do not share the same support merely because they appear on one globe. A compact country symbol also cannot preserve every internal spatial difference.

Small territories and islands may have little or no representation in a sampled spatial layer. Country rosters differ between sources and can include territories or economies in different ways. A count of economic entries therefore cannot be substituted for the number of emotional country observations. An absent mark can arise from source coverage, geographic assignment, or representation at the selected scale.

Interaction changes attention and sometimes the comparison

Rotation and zoom change the viewer's position. Selecting a country or category brings a part of the work into focus. Restricting emotional categories changes the set from which the visible winner is chosen: a different word can appear without any underlying article changing.

Animated connections organize attention from a selected starting place. They do not provide temporal evidence of distress spreading. The soundtrack is composed; interaction tones are artistic responses to selected signals. Neither is a recording of a population's suffering.

The related web experience can allow optional personal contributions. This is separate from the country datasets and the large PCAI analysis. Projection mode emphasizes the shared map and hides the survey action. Personal disclosure is not necessary to understand the work.

Source scope: what has actually been counted

The following distinctions apply to every source: its original extent, the selected material analysed, the observations represented, geographic coverage, language metadata, and time. An institution's entire archive is not equivalent to the project's selection from it.

Physical, environmental, and economic sources

Source family	Selected exhibition material	Scope and unresolved details
IHME Global Burden of Disease with WorldPop	15,282 mapped condition observations, 2,547 for each of six categories.	Health estimates combined with population information; original health measure, reference year, and selected release remain unconfirmed.
Berkeley Earth	8,100 temperature-change observations.	Spatial temperature information; the selected reference climate, comparison period, and transformation remain unconfirmed.
Climate TRACE	2,339 emissions-related observations.	Activity-related emissions information; selected release, gas definition, period, and transformation remain unconfirmed.
World Bank GDP per capita	190 available 2024 values and 16 missing entries in the exhibition roster.	Current US dollars per person; economic entries do not necessarily match the emotional-country roster.

The physical count is an output count of mapped observations, not the number of people in the original health studies. Likewise, 8,100 temperature points do not establish 8,100 weather stations, and 2,339 emissions points do not establish 2,339 unique facilities. Exact upstream record counts for these selected health and environmental releases are not established here. The global source programmes contain much more material than these exhibition selections. [\[S7-S11\]](#)

Country summaries contain 13,183 physical condition-points across 155 represented countries, 2,591 temperature points across 140, and 1,873 emissions points across 146. The remaining 2,099 physical, 5,509 temperature, and 466 emissions points are layer inputs without assignment to the selected country polygons. These counts distinguish an input layer's extent from the support of its country summaries; they do not mean those remaining inputs are deleted.

WorldPop's cited product family covers 2000-2020 and includes UN-adjusted population estimates at approximately 1 km resolution. That identifies the population source family; it does not establish the exact year used in this exhibition's placement. Language metadata is relevant to textual sources, while these numerical layers have units, periods, and spatial support rather than a sampled population of spoken languages. [\[S8\]](#)

Why incomplete source details remain visible

The guide can explain what a layer represents and compare its selected relative signals. It cannot reconstruct an original clinical rate, temperature anomaly in degrees, or emissions total in tonnes from appearance alone. Such conversions need the specific source measure, reference period, geographic support, and transformation. These unresolved items are stated at the relevant layer, without inventing a measurement history from the source institution's general documentation.

Text-source counts and their original scope receive fuller treatment below because PCAI's preparation and interpretation are central research components of the project.

Physical pain: burden, concentration, and aggregation

The physical layer makes the consequences of painful conditions visible through a marked surface. It connects country-based health information with population information from WorldPop. Population concentration is consequently relevant to strong local signals, including India's.

Six related but distinct conditions

Category	Conceptual distinction
Headache disorders	A broad group of recurring headache conditions; a source of disability in daily activity.
Migraine	A neurological condition within the broader headache domain, often with symptoms beyond headache.
Low back pain	Pain in the lower back, with potential restrictions on mobility and participation.
Neck pain	Pain affecting the neck and activities that depend on comfortable movement or posture.
Osteoarthritis	Joint disease associated with structural changes, pain, stiffness, age, and other risk factors.
Rheumatoid arthritis	Immune-related inflammatory disease, distinct from osteoarthritis.

The categories overlap conceptually and can coexist within a person. Adding migraine to the broader headache category would not yield a defensible count of different patients. General clinical descriptions clarify the topic, while the exhibition's original health measure remains unconfirmed. It should not be silently identified as prevalence, incidence, or disability-adjusted life years. [C3, C5-C7]

Counts, rates, and concentration

A **count** concerns how many events or people are represented; a **rate** relates events to a population and usually a period. A **burden measure** may instead weight consequences, such as activity limitation. These quantities can order countries differently. A populous location can carry a large aggregate burden even when its residents do not have the highest individual risk. The population-based spatial placement makes that distinction especially relevant here.

The compact country summary emphasizes a maximum local signal. It answers a question about the strongest represented location, rather than the average condition throughout a country. If the spatial signals are v_1 through v_n , this summary is conceptually $\max(v_1, \dots, v_n)$. It is not their arithmetic mean. A country with more observed locations also has more opportunities to contain an extreme value.

This is an explanatory property of the representation, not evidence that dense populations cause a painful condition. Age structure, health risks, health care, reporting, and source modelling might contribute to real-world patterns. Their separate contributions to this exhibition signal have not been estimated.

The artistic question remains substantive: what becomes perceptible when limits on movement, sleep, concentration, and participation are given an external form? One possible framing is: "The surface brings a burden into view that often remains invisible in everyday encounters."

Environmental pain: two quantities in one atmosphere

The environmental layer combines temperature change and emissions-related pressure in a shared atmospheric form. Their relationship is scientifically meaningful, but their measurement units and interpretation remain different.

Temperature change and absolute heat

A temperature anomaly expresses a difference from a reference climate. A warming comparison therefore needs both the period of interest and its reference period. A location may remain cold while warming substantially; another may be hot while showing a smaller change relative to its own reference. Russia and Canada make this distinction accessible in the exhibition.

Arctic amplification provides established regional context: changes in snow and ice reflectivity interact with atmospheric and oceanic processes to produce strong high-latitude warming. It does not supply a complete explanation of each displayed location. The selected Berkeley Earth comparison period and baseline are still unresolved, so the image should not be translated into a specific temperature increase. [S9, C1]

Emissions, concentrations, exposure, and vulnerability

Emissions describe releases associated with activities. **Concentration** concerns the amount of a substance present in an environment. **Exposure** concerns contact with a person or system. **Vulnerability** concerns susceptibility to harm. A high value in one concept does not supply the missing measurements in the others.

Climate TRACE provides estimated emissions across activities and gases. Carbon dioxide and carbon-dioxide-equivalent totals also differ: the latter combine gases using a specified climate equivalence framework. The exact gas measure and period selected for the exhibition remain unconfirmed. No visual conversion into tonnes, measured smoke, or inhaled pollution is justified. [S10]

Local extremes and shared processes

The country summary emphasizes a strong local signal. Russia's or Iran's strong emissions mark therefore does not establish a ranking of national emission totals or emissions per person. Uruguay's observed zero in the temperature-change view is a fact about this selected transformed signal, not evidence of no climate change or environmental vulnerability.

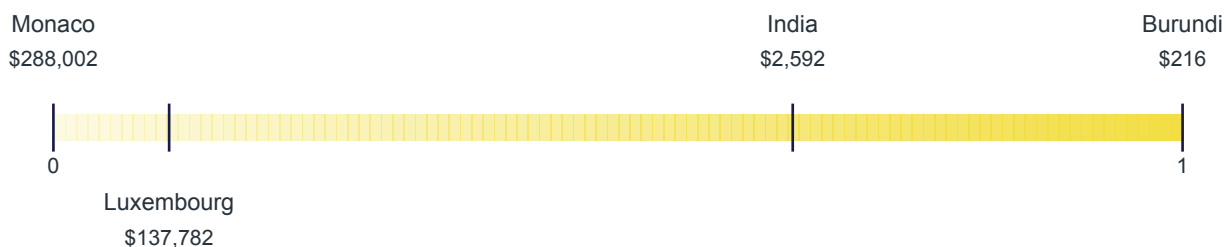
Artistically, an atmosphere that extends beyond borders places local activities within shared conditions. Scientifically, the image invites questions about transport, accumulation, scale, and responsibility without having measured those relationships as network edges. Warming and emissions can be considered together while retaining the different questions their indicators answer.

Socioeconomic pain: an economic proxy and its denominator

The yellow layer uses World Bank GDP per capita for 2024, measured in current US dollars. GDP is production within an economy; dividing by population provides an output-per-person indicator. Lower values receive stronger yellow. This introduces material circumstances into the artwork while leaving distribution and individual experience unresolved. [S11]

Why the logarithmic display matters

Economic output spans a large range across the selected economies. A logarithmic display compresses that range so values at its lower end remain visually distinguishable. Equal distances on a logarithmic axis correspond to equal ratios, rather than equal dollar differences. The display reverses direction so lower output per person produces a stronger hardship proxy.



GDP per person decreases as the artistic yellow signal increases.

This transformation does not convert the measure into purchasing-power parity, correct for inequality, or estimate household poverty. Current US dollars also retain currency-conversion effects. A familiar numerical unit can make a comparison look more comprehensive than it is: GDP per person remains different from disposable income, median income, wealth, and access to public services.

Economic structure changes the meaning of an average

Luxembourg illustrates a denominator issue. Cross-border workers contribute to domestic production while residing outside the population denominator. Its high GDP per person cannot be interpreted as the typical salary of a resident. Burundi's low value invites attention to economic constraints, but the map alone cannot identify their causal shares or describe how every household lives. Monaco's weak yellow signal does not make emotional or bodily pain logically inconsistent. [C2, C4]

There are 190 available observations and 16 missing entries in the exhibition's economic roster. Missingness, including Yemen's selected 2024 value, is a property of this observation set. Replacing a missing value with zero would create an artificial economic extreme. Conversely, treating an absent yellow mark as prosperity would reverse the evidential meaning of the blank.

Emotional categories, language, and the observer

The emotional layer presents distress categories inferred from texts. Some passages express an author's experience; others report, quote, dramatize, or discuss somebody else's experience. Recognizing distress in language does not identify whose emotion it is, how enduring it is, or how common it is among residents of the associated country.

Fourteen broad openings into the text

Display category	Scope of meaning
Body pain	Language expressing bodily hurt.
Hurt	Emotional suffering without a more specific assigned form.
Eco-anxiety	Fear or anxiety connected with environmental change.
Uncertainty	Distress associated with instability and an unclear future.
Grief	Loss and mourning, including distress over unwanted change to a home environment.
Anger	Anger, outrage, and moral injury associated with perceived wrongs.
Hardship	Loss, insecurity, and difficult material circumstances.
Displacement	Distress associated with exile, migration, or being uprooted.
Trauma	Language of trauma and overwhelming experience.
Loneliness	Social and relationship pain.
Depression	Depressive distress; the research taxonomy includes more severe forms.
Fear	General fear, anxiety, and panic.
Helplessness	Perceived inability to influence circumstances.
Shame	Shame, guilt, and self-blame.

These categories are an interpretive taxonomy, meaning an organized set of distinctions. They are neither an exhaustive theory of emotion nor diagnostic assessments of populations. Several categories may apply simultaneously. The display's single prominent word is a reduction of that multidimensional representation.

The 47 finer distinctions

The taxonomy refines the broad groups as follows. These labels describe what the classifier is trained to recognize in language, not a set of independently diagnosed conditions.

Broad group	Fine categories
Body pain	Acute physical pain; chronic physical pain; somatic pain distress; physical exhaustion.
Hurt	Sadness; general emotional distress.

Broad group	Fine categories
Eco-anxiety	Climate fear; climate anxiety; climate panic.
Uncertainty	Future uncertainty; situational instability; loss of control.
Grief	Bereavement grief; grief over loss other than death; solastalgia; ecological or climate grief.
Anger	Anger; injustice anger; betrayal distress; moral injury.
Hardship	Material loss; financial or livelihood precarity; food insecurity; housing insecurity.
Displacement	Forced displacement; exile; loss of home or rootlessness; climate or ecological displacement.
Trauma	Traumatic stress; hypervigilance; acute overwhelm; shutdown or dissociation.
Loneliness	Loneliness; social isolation; abandonment; rejection or relationship loss.
Depression	Depression; hopelessness; anhedonia or emotional numbness.
Fear	Fear; anxiety; panic.
Helplessness	Helplessness; powerlessness.
Shame	Shame; guilt; self-blame.

In this taxonomy, **moral injury** is distress associated with perpetrating, witnessing, or being betrayed by moral violations. It is more specific than ordinary anger at unfairness. **Solastalgia** concerns a changing home environment while a person remains there, distinct from displacement. **Anhedonia** means loss of interest or pleasure; **hypervigilance** means heightened monitoring for threat. These concepts help readers distinguish neighbouring labels.

Separate safety outputs concern suicidal-ideation language and self-harm language. They do not assess intent, planning, imminence, or clinical risk. Four out-of-scope outputs distinguish unsupported content types, unclassifiable noise, insufficient semantic content, and content outside the distress domain. A derived neutral/no-pain control is not a 48th learned distress category. These distinctions explain why 70 model outputs do not mean 70 emotional themes.

Three language questions

The language of an article, a language hint supplied with a record, and the language chosen for a displayed word are different things. A display translation creates access and resonance; it does not establish that the underlying country collection was written in that language. The 192 displayed country labels use 101 language codes and 20 scripts from a curated lexicon. This is separate from the 100-language training roster. One selected display language cannot represent every linguistic community within a country.

Translation between brief category names is not guaranteed to preserve every cultural or emotional association. A shared category is useful for comparison, while **measurement invariance**, comparable meaning and performance across groups, would require separate evidence. Neither multilingual training coverage nor a multilingual globe demonstrates it.

An optional formulation is: "A shared word creates a point of recognition across difference; it does not erase the differences in whose language and experience reached the map."

PCAI: the adopted classifier and what its scores mean

PCAI prepares country-associated texts; the classifier interprets them through a taxonomy of physical, emotional, and social distress. The adopted version 9 was finalized on 4 September 2026. It combines a frozen pretrained BGE-M3 multilingual encoder with a project-trained neural classifier. The encoder is not trained from scratch on the PCAI examples. [S12]

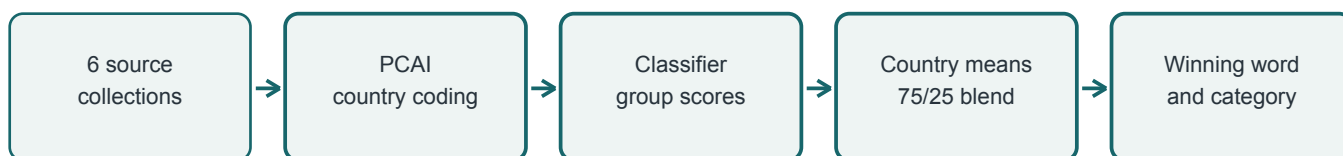
The classifier is designed to classify texts in 100 languages, and its synthetic teaching collection is multilingual across those languages. That coverage is separate from demonstrated accuracy: the research does not establish equally reliable performance in all 100 languages.

Representation learning and the project-specific classifier

An **embedding** is a numerical representation of a text. BGE-M3 produces a 1,024-dimensional representation; the classifier learns associations between these representations and the project's labels. The encoder remains fixed while the project-specific network learns. The adopted network has a shared hidden layer of 1,024 units with a smooth nonlinear activation, followed by label-specific outputs. A shared layer allows related outputs to depend on a common transformation of the text representation.

There are 70 outputs: three scope decisions, 14 broad groups, 47 fine categories, two safety-related signals, and four reasons for being outside the task. These are not 70 emotions. The safety outputs concern language signals, not clinical assessment. Fine categories refine broader ones, allowing a passage to express more than one specific form of distress.

The classifier sees at most 512 tokens per record. Tokens are text fragments rather than a fixed number of words. BGE-M3 has a broader underlying capacity, but the adopted PCAI analysis uses this shorter view. Long articles can therefore lose later qualifications, quotations, or shifts in perspective. The share affected depends on genre and language. In the earlier 5,000-news diagnostic selection, 2,479 records exceeded this limit; that diagnostic proportion should not be assigned to the entire main corpus.



Training examples are separate from the real texts analysed here.
Earlier exhibition pipeline; the latest analysis uses equal record weights.

Scope and hierarchical composition

The model first provides scores for distress being present, no relevant pain, and content outside the task. The three scope scores are normalized to sum to one. Broad category scores are conditional signals multiplied by the normalized distress score. Fine-category scores are then multiplied by the score of their parent broad category.

For text x and category k , the conceptual relationship is:

Broad score k = normalized distress score $\times$ conditional category score k . Fine score = parent broad score $\times$ conditional fine-category score.

This hierarchy prevents a fine category from exceeding its parent in the returned score structure. It also means a scope error can affect every descendant score. A model that overassigns distress relevance can raise many downstream scores even when the finer distinctions are otherwise useful. The synthetic training material is distress-heavy, making transfer to general news a substantive concern.

The outputs are probability-like scores. Hierarchical multiplication is a modelling decision, not proof of probabilistic calibration. Calibration would mean that, among comparable texts given a particular probability, the corresponding human-labelled frequency agrees with it. That property must be checked in the target domain.

Scores, decisions, and visible winners

A threshold turns a continuous score into a positive decision. Scope decisions additionally use a required separation between the leading alternatives; an uncertain result is possible. The project's diagnostic news decoding uses these threshold and margin rules, rather than treating the largest scope score as automatically sufficient.

Three summaries must therefore be distinguished:

- A **mean score** averages the continuous model outputs across texts.
- A **positive fraction** counts texts crossing a specified decision threshold.
- A **displayed winner** is the largest permitted country-level category score.

The display excludes no-pain and out-of-scope controls when choosing a word. At least one control exceeds the winning pain score in 121 of the 192 current country results. Display prominence therefore does not certify that distress dominates the texts.

The emotional country values used here are mean scores. A value of 0.25 is not a finding that 25 percent of residents are distressed, nor necessarily that 25 percent of articles crossed a positive threshold. A single winning word can conceal several other substantial scores. Conversely, even a low set of scores has a largest member. The 14 broad scores are not normalized to sum to one, because several categories can apply.

PCAI training, evaluation, and limits of generalization

The teaching material is synthetic, constructed to express known kinds of distress across varied languages and forms. Synthetic examples permit deliberate coverage of labels and combinations, but their known intended labels do not make them equivalent to naturally occurring testimony. The real news, posts, and complaints analysed afterward are a separate resource.

Prepared material and the material actually fitted

Resource	Size	Role
Prepared synthetic collection	573,143 examples across 100 languages.	Broader teaching resource.
Adopted fitted subset	333,236 examples.	Material used to fit the current classifier.
Current-generation fitted component	295,572 examples.	Main component of the fitted subset.
Earlier reviewed fitted component	37,664 examples.	Complementary earlier material.
Clean synthetic reporting test	15,874 examples.	Withheld from fitting, subsequently consulted during model comparison.
Withheld synthetic stress half	10,429 examples.	Challenging material used as an adoption check after selection.

The two fitted components sum to 333,236. Older mapped-label material was excluded from the adopted recipe. The broader collection, fitted subset, and evaluation resources describe different stages and memberships; their counts should not be added into a training total. The 100-language figure describes coverage, not equal sample sizes or demonstrated accuracy for each language.

The 100 languages

Languages 1-25	Languages 26-50	Languages 51-75	Languages 76-100
Afrikaans	Galician	Latvian	Sinhala
Albanian	Georgian	Lingala	Slovak
Amharic	German	Lithuanian	Slovenian
Arabic	Greek	Luxembourgish	Somali
Armenian	Guarani	Macedonian	Spanish
Assamese	Gujarati	Malagasy	Sundanese
Azerbaijani	Haitian Creole	Malay	Swahili
Basque	Hausa	Malayalam	Swedish
Belarusian	Hebrew	Maltese	Tajik
Bengali	Hindi	Marathi	Tamil
Bosnian	Hungarian	Mongolian	Telugu
Bulgarian	Icelandic	Nepali	Thai
Burmese	Igbo	Norwegian Bokmål	Tigrinya
Catalan	Indonesian	Odia	Turkish
Cebuano	Irish	Oromo	Turkmen
Chinese	Italian	Pashto	Ukrainian
Croatian	Japanese	Persian	Urdu
Czech	Javanese	Polish	Uyghur
Danish	Kannada	Portuguese	Uzbek
Dutch	Kazakh	Punjabi	Vietnamese
English	Khmer	Quechua	Welsh
Estonian	Kinyarwanda	Romanian	Xhosa
Filipino	Korean	Russian	Yiddish
Finnish	Kyrgyz	Serbian	Yoruba
French	Lao	Shona	Zulu

These are the language entries in the multilingual training roster and intended classifier coverage. They do not establish equal accuracy or coverage of every dialect, script, and mixed-language form. Norwegian Bokmål is the Norwegian entry; Chinese and Arabic each count as one broad entry.

What the evaluation numbers establish

Evaluation	Broad-category macro-F1	Fine-category macro-F1
Clean synthetic reporting test	0.9028	0.7772
Withheld synthetic stress half	0.6658	0.6552

Precision is the fraction of predicted positives that are labelled positive; **recall** is the fraction of labelled positives that the model retrieves. F1 is their harmonic mean: **$F1 = 2 \times \text{precision} \times \text{recall} / (\text{precision} + \text{recall})$** . Macro-F1 gives each evaluated category equal weight. It therefore answers a different question from a record-weighted accuracy score or a measure dominated by common labels.

The lower stress results expose difficulty under more challenging combinations and prevalence patterns. The stress set was deliberately constructed, so it is not a representative estimate of the error rate in world news. Nor does a broad score near 0.90 establish that the classifier is 90 percent correct about residents' feelings.

The training work also examines the Brier score, the mean squared difference between a predicted probability and a binary reference label. It evaluates probability predictions, whereas F1 depends on a decision threshold. Brier scores are affected by label prevalence; a stress set with a different base rate needs appropriate comparison. Discrimination, calibration, and thresholded performance can improve or deteriorate separately.

Independence has more than one meaning

The clean split groups related source identities to reduce leakage of closely related material between fitting and evaluation. That protects against a specific duplicate-related failure. It does not make all examples semantically independent or eliminate shared generator style.

The clean reporting test was consulted repeatedly during development. It was withheld from parameter fitting, but it was not an untouched final assessment of the entire model-selection process. A separate stress half was reserved as a veto after candidate selection. The broader stress resource had itself been consulted earlier in development; splitting it later does not erase that history. The half gives additional evidence, rather than a pristine external test. Its split also lacks the source-group protection available for the clean data, so related examples can cross the two stress halves.

Transfer to real-world language

Domain shift means that deployment material differs from the training and evaluation material. News reporting differs from constructed first-person distress through quotation, negation, third-person narration, mixed topics, and uncertain speaker identity. Language coverage may differ too. Aggregate country comparisons can inherit these changes.

The adopted classifier is used for project research; independent real-world validation remains unestablished. The completed large-scale run demonstrates that it produced results for the selected texts. It does not establish real-world semantic accuracy, clinical validity, or national emotional prevalence. Stronger claims would require independently labelled real material across sources, languages, periods, and relevant forms, with disagreement and uncertainty reported. Synthetic success and large inference volume answer useful but different questions.

CCNews: from approximately 600 million to the main corpus

CCNews supplies general international news gathered through Common Crawl and prepared by Stanford OVAL. The upstream description gives approximately 600 million articles in more than 100 languages. It already includes cleaning, deduplication, and language identification. Its crawl-period description runs from 2016 through June 2024 and distinguishes crawl dates from article publication dates. [S1]

PCAI's locally acquired snapshot contains 474,985,436 records. The approximate upstream figure and exact local figure do not establish a precise count removed by the project: they describe different scopes. Further local preparation and country selection produce the main 393,745,301-record corpus across 186 publisher countries.

The counted stages

Stage	Records	Interpretation
Upstream source description	About 600 million	Approximate extent of the broader Stanford OVAL resource.
Locally acquired snapshot	474,985,436	Actual local acquisition count.
Prepared text collection	472,438,058	After initial duplicate/date checks and later privacy cleaning.
Supported publisher-country attribution	394,231,021	Usable and non-conflicting geographic assignment.
Main country-linked corpus	393,745,301	After final eligibility and duplicate checks.
Recorded-year 2015-2026 subset	388,842,327	Nested within the main corpus.

The main corpus also contains 4,902,974 records with recorded years 2010-2014. **388,842,327 + 4,902,974 = 393,745,301**. These are date partitions of one collection, not separate datasets to be combined again. Both the main corpus and its later-year subset are reasonably described as roughly 400 million when the precise scope is also stated.

Filtering and the arithmetic of exclusion

Initial normalization standardizes characters, whitespace, and escaped symbols so equivalent text can be compared. Initial checks remove 906,602 duplicates and 1,640,769 records failing supported date rules, leaving 472,438,065. Later privacy cleaning reveals seven additional duplicate collisions, giving 472,438,058 prepared records.

Country assignment excludes 78,207,037 records: 78,030,618 unresolved assignments and 176,419 conflicts. This retains 394,231,021. Text, privacy, and record eligibility checks remove 485,075 more, followed by 645 final duplicate or conflict exclusions. The result is 393,745,301. These stages explain the exact local reduction without treating every removed record as a duplicate or a privacy problem.

Privacy preparation masks detected email addresses and telephone numbers; later checks also mask links and user handles. Masking usually changes a text rather than discarding it. It reduces recoverable identifiers but

does not prove complete anonymization. Eligibility also requires adequate readable prose and consistent record information. The combined eligibility count cannot be interpreted as a count of harmful articles.

Selection creates a particular view of the world

The collection is general news. It does not require a pain keyword for entry; the classifier assesses distress relevance afterward. The main corpus has no per-country cap. Its distribution reflects publisher output, web-crawler reach, language availability, and successful country resolution. Removing unresolved publishers can alter representation even when the assignment rule is applied consistently.

The acquired and prepared collections contain 116 recorded language codes. That figure differs from the classifier programme's 100 training languages and the completed news selection's 104 language or variety hints. These describe different inventories and cannot be exchanged.

Recorded dates can derive from article metadata, crawl information, or an acquisition-time substitution when another usable date is absent. A recorded 2026 year does not prove that an article reports a 2026 event. The main dated collection contains no 2025 entries. Initial date eligibility was not simply a 2015-2026 filter, as the retained earlier records show.

Prepared does not mean completely classified

The main corpus supports the broader ongoing classification programme. As of the project records checked on 9 September 2026, a full 393.7-million-record production classification run is not established as started. The verified completed news analysis covers 25,932,030 selected records within that corpus. No full-corpus completion percentage is claimed here.

The latest completed analysis and its six sources

The latest news run was completed and summarized by 8 September 2026. It applied the adopted classifier to 25,932,030 articles and combined their results with 244,438 records from five smaller sources. The total is 26,176,468 records across 192 observed countries. It was an application of the model to real material, rather than a new training run.

News and expression sources

Source	Records analysed	Countries represented
Stanford OVAL CCNews	25,932,030	186
Twitter Sentiment by Country	166,837	163
DEMOTEC participatory-budgeting tweets	67,193	136
Yachay coordinate-linked texts	8,564	4
CFPB consumer complaints	1,030	1
CMU GeoText microblogs	814	1
Combined	26,176,468	192

Country coverage overlaps, so the country counts do not sum to the combined total. The five smaller sources together cover 175 countries. Records are not unique people, and multiple records can concern the same event or author.

Source-specific scope and selection

CCNews: The main and upstream counts are given in the preceding chapter. The completed selection sought up to 250,000 articles per country with attention to recorded-year coverage; 86 countries reached that target. It contains 104 language or variety hints. Approximately 94.2 percent carries recorded years 2017-2024. Publisher country is organizational origin, not event location, author nationality, or audience residence. The displayed earlier selection uses 5,000 news articles. [\[S1\]](#)

Twitter Sentiment by Country: The analysis includes 166,837 social-media records associated with 163 countries and recorded year 2020. The selection has 60 language hints; country context comes from source-provided metadata. The same selected material contributes to the earlier exhibition analysis. An exact original release size and complete upstream language inventory are not established here; the analysed count should not be labelled the size of all available Twitter material. [\[S2\]](#)

DEMOTEC: The analysis includes 67,193 tweets concerning participatory budgeting and civic participation, associated with 136 countries and recorded years 2015-2021. Its analyzed selection carries nine language hints. The source's topic focus and profile-location geography shape the resulting view of public expression. The original release's exact total is not established here. The displayed earlier analysis uses this same 67,193-record selection. [\[S3\]](#)

Yachay: The source describes more than 600,000 coordinate-linked posts focused on six countries. PCAI analyses 8,564 selected records across four country contexts, with recorded years 2014-2022. The same selection contributes to the exhibition. Coordinate context is geographic evidence, without proving nationality or a stable place of residence. The selected records carry 47 language hints. [\[S4\]](#)

CFPB: The analysis includes 1,030 consumer complaints in one country context, with recorded year 2025. The complaint database is an evolving institutional collection; its total size at the project's acquisition snapshot is not established here. Complaints solicit accounts of problems, a different selection mechanism from general news or ordinary conversation. The same subset is used in the exhibition. The selection carries two language hints. [S5]

CMU GeoText: The analysis includes 814 geographically associated microblog records in one country context, with recorded year 2010. The cited corpus release dates to October 2010; its full original count is not established in this guide. The same 814-record subset contributes to the exhibition. This older geographic-language resource is not a current survey, and the selection carries eight language hints. [S6]

Uneven language coverage within broad country coverage

Recorded news hint	Articles
English	6,335,174
Spanish	4,489,844
Arabic	2,268,925
French	1,658,090
Russian	1,283,339

The selection includes country-associated news for China and Japan but very few records identified as Chinese or Japanese. Broad country coverage therefore does not establish representative local-language coverage. Nor is a language hint a guarantee of correct identification, translation equivalence, or classifier quality in that language.

Aggregation: which question does a country score answer?

Aggregation defines the population of records being summarized and their influence. Changing the weighting changes the question, even when the classifier remains the same.

The current exhibition and the latest research differ

The current exhibition uses 249,438 texts: 5,000 news articles plus 244,438 expression records. Its news has 59 language hints; 93.84 percent has recorded years 2015-2016. The combined exhibition texts have 84 hints and discontinuous recorded years 2010-2025. Its difference from newer research therefore includes source dates as well as size and weighting. Where both source families are available, it gives the news-family average 75 percent of the weight and the expression-family average 25 percent. Where only one family is available, the available family supplies the result.

Exhibition score = $0.75 \times \text{news mean} + 0.25 \times \text{expression mean}$, where both exist.

The latest completed analysis instead gives one equal contribution to each available record. Its country score is the sum of the selected record scores divided by that country's record count. Equivalently, each source-family mean is weighted by its actual share of records.

For example, India's displayed selection has 27 news and 1,442 expression records. Its news mean still receives 75 percent of the blend, showing why a record count alone cannot describe a source's influence. The later much larger selection uses a different rule.

Latest score = $(\text{news count} \times \text{news mean} + \text{expression count} \times \text{expression mean}) / \text{total count}$.

The same texts can support different summaries

EXHIBITION

$$0.75 \times \text{news mean} + 0.25 \times \text{expression mean}$$

Fixed source-family weights when both exist

LATEST COMPLETED ANALYSIS

$$(\text{news score sum} + \text{expression score sum}) / \text{all records}$$

Each record has equal weight within its country

Because news supplies 25,932,030 of the 26,176,468 combined records, its global share is 99.066192 percent, approximately 99.07 percent. This is not every country's news weight; source composition varies by country. Equal treatment of records is different from equal treatment of publishers, people, languages, countries, or source families.

Winners and counterfactual views

With all broad categories available, anger/moral injury leads in 152 of 192 exhibition country results. In the latest combined research it leads in 178 of 192, compared with 174 of 186 news-only results and 124 of 175 expression-only results. Those counts concern the strongest model score, not a majority vote of residents or a demonstrated prevalence.

Excluding anger lets the next-largest permitted score become visible. It answers which theme would lead under a restricted comparison; it does not establish that anger is absent. A winner can also change when two nearly tied scores exchange order. The visible label alone does not display that margin or a confidence interval.

Latest examples with their denominators

Country	Text records	Highest broad mean score
India	251,442	Anger/moral injury: 0.2466.
Japan	251,096	Anger/moral injury: 0.1752.
United States	258,375	Anger/moral injury: 0.1698.

India's total includes 250,000 news articles and 1,442 expression records. The United States includes additional complaint and microblog sources, giving it a different mixture. Countries with other latest leading themes include Iceland (unspecified emotional pain), Saint Vincent and the Grenadines (grief), Tonga (fear), and Samoa (hardship). These are newer results, not replacement descriptions of the earlier projected labels.

Comoros, Micronesia, and Guinea-Bissau have no emotional observations in the completed combined inputs. A missing entry must remain distinct from a low or zero model score.

Delaunay and Gabriel: geometry within a shared category



The illustrated anger network identifies the active category, geographic connections, and a country in focus. Countries enter a category's network through the displayed emotional winner. The edges are then constructed from geographic positions. This separates the semantic criterion for belonging to a group from the spatial criterion for connecting its members.

What a graph means here

A **graph** contains nodes and edges. Here the nodes stand for country locations and edges organize proximity within a shared category. An edge is not a measured correlation between time series, an inferred causal effect, a social tie, or evidence of psychological transmission. Two countries can share a category without sharing a history or having a direct connection.

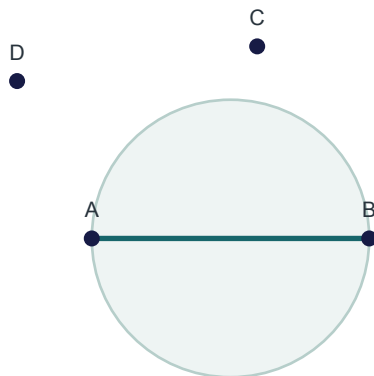
The current exhibition uses a spherical Delaunay construction. Its positions are chosen country label locations, with adjustments for small places, rather than mathematically calculated country centroids. Conceptually, its triangular mesh can be obtained from the convex hull of the category's positions in three dimensions. The **convex hull** is the tight outer shell around a set of points. Its triangular faces provide neighbouring connections on the sphere. The use of a globe avoids treating longitude as an ordinary flat coordinate with an artificial break at the map's edge.

Gabriel as the requested comparison

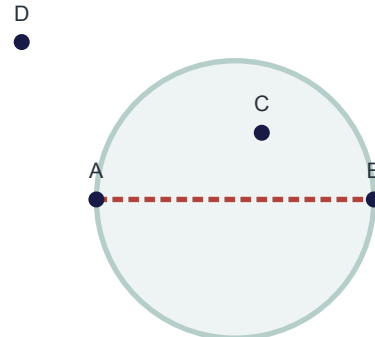
A Gabriel connection uses an empty-region test. For two positions A and B, form a ball whose diameter is the straight segment joining them. Their direct edge is retained only when no other participating position lies inside that ball. In a flat diagram the ball appears as a circle. The spherical version tests positions embedded in three dimensions, with chord geometry defining the ball.

If M is the midpoint of A and B, the interior test asks whether another position C has $\text{distance}(C, M)$ smaller than $\text{distance}(A, B)/2$. A nearby point in that interior makes the direct A-B connection unnecessary under the Gabriel rule. This is a geometric definition, not a judgement about how similar the countries' emotions are.

The idea behind a Gabriel connection



KEEP: no point inside



REJECT: C is inside

A flat illustration of proximity; the globe uses the same idea in three dimensions.

Delaunay generally supplies a denser proximity mesh; Gabriel supplies a more selective one. For the current all-category view, anger has 152 nodes and 450 Delaunay edges; the compared Gabriel construction retains 279. Hurt has 18 nodes and 48 versus 30 edges; hardship has six nodes and 12 versus seven. These are construction counts, not relationship strengths.

The project's small-set rule matters: a three-country category uses all three pairwise edges in either mode, without applying the Gabriel interior test. A one-country category has no edge. Thus the mathematical Gabriel definition and the project's small-set treatment should not be conflated. The alternative is shown here for comparison; Delaunay remains the current exhibition network.

The empty-region definition depends on which positions participate in the comparison. Changing the visible category membership can therefore change geometric connections even when the country locations remain fixed. Curved drawing and animation make the connections legible on the globe; they do not add evidence of transmission, direction, speed, or chronology.

Country stories / 25 evidence-based entry points

The following cards concern the current exhibition unless explicitly marked as latest research. Each observation belongs to a named metric and its available support. Context gives an independently supported explanation of a relevant concept; possible interpretations remain questions where the display has not estimated a causal mechanism. High and low never mean an overall ranking of whose suffering matters.

Physical and environmental values below are **unitless exhibition signals**, not clinical rates, degrees, or tonnes. Their n counts assigned map observations. For physical categories, one observation represents a condition at a location, not a unique person. Emotional n counts texts. GDP retains its original current-US-dollar unit. Ranks concern only the countries with an observation for that metric. Decimal precision does not imply equivalent measurement accuracy.

Story 01 / India: a concentrated burden

Observation: Migraine maximum 0.94266 from 45 mapped migraine observations, third among 148 observed countries. The overall physical view has 266 assigned condition-points.

Interpretation: Population-based placement helps explain concentrated burden where many people live. It does not establish unusually high risk for each resident. Recurring pain's effects on study, work, and care make the collective signal meaningful without turning it into a uniform description of India.

Story 02 / Pakistan: headache burden and overlapping categories

Observation: Headache and migraine maxima both reach 1.00000, with 13 observations each. They lead their respective comparisons of 149 and 148 observed countries.

Context: WHO recognizes headache disorders as a substantial cause of disability. [C7]

Interpretation: The categories can illuminate repeated interruption of everyday activity. Their overlapping clinical meanings prevent adding these marks into a count of separate patients. A maximum is also different from national prevalence.

Story 03 / Japan: mobility and ageing

Observation: Back-pain maximum 1.00000 from 17 observations, highest among 150 observed countries.

Context: Japan's population aged 65 or over was 29.3 percent in 2024. Age is relevant to the wider burden of back pain. [C3, C8]

Interpretation: Ageing offers a plausible contextual lens on mobility and independence. It is not a measured decomposition of this spatial maximum. Living longer and maintaining participation are connected but distinct concerns.

Story 04 / China: two kinds of arthritis

Observation: Osteoarthritis and rheumatoid-arthritis maxima each reach 1.00000, from 135 and 137 observations respectively, leading their comparisons of 149 and 153 countries.

Context: Osteoarthritis and immune-related rheumatoid arthritis involve different clinical processes. [C5-C6]

Interpretation: Similar effects on movement can arise through different mechanisms. The artwork opens a discussion of daily function; the country signal cannot diagnose individuals or assign a mechanism to their pain.

Story 05 / Egypt: movement and the temptation to invent a cause

Observation: Neck-pain maximum 0.99900 from 12 observations, highest among 148 observed countries.

Interpretation: Reading, turning the head, and moving through a day make neck pain a recognizable topic. The selected data do not establish an occupational, postural, or lifestyle cause for Egypt's strong mark. A familiar possible cause is not automatically the explanation of a country maximum.

Story 06 / United Kingdom: invisible limitations

Observation: Rheumatoid-arthritis maximum 0.99961 from nine observations, second among 153 observed countries.

Context: Immune-related inflammation can affect joints and other tissues. [C6]

Interpretation: The scarred surface makes room for bodily limitation that may be invisible to an observer. The nearly maximal number concerns one represented local signal, not the fraction of people affected or the severity experienced by a typical resident.

Story 07 / Guyana and Vanuatu: low signals with sparse support

Observation: Overall physical maxima are 0.00073 for Guyana and 0.00202 for Vanuatu, second- and third-lowest among 155 observed countries. Each has 12 condition-points, two per condition.

Interpretation: A quiet appearance needs the same attention to support as a dramatic one. Limited mapped observations and a selected condition inventory leave substantial experience outside the image. Low signal is a bounded finding, not a conclusion that residents seldom experience pain.

Story 08 / Russia: strong change in a cold environment

Observation: Temperature-change maximum 0.7058875 from 363 observations, highest among 140 observed countries.

Context: Arctic amplification explains why cold high-latitude regions can change strongly. [C1]

Interpretation: Coldness and warming answer different questions. The local maximum neither estimates Russia's average warming nor supplies the unresolved source baseline. Regional climate science helps explain the conceptual distinction without reconstructing the selected exhibition measure.

Story 09 / Canada: a local extreme within a large country

Observation: Temperature-change maximum 0.614436875 from 220 observations, second among 140 observed countries.

Interpretation: A large national outline contains varied environments. The maximum directs attention to a strong represented change, while leaving its spatial distribution unresolved. Arctic warming provides relevant regional context, but a national maximum cannot describe every community's climate history or vulnerability. [C1]

Story 10 / Uruguay: an observed zero

Observation: Temperature-change maximum 0 from two observations, tied at the minimum with Timor-Leste's single observation.

Interpretation: This is a zero in the transformed exhibition signal with observed support. It differs from missing information. Neither case establishes no climate change. Uruguay illustrates why a source value, an absent observation, and an environmental conclusion must remain distinct, even when their visual treatments look similarly quiet.

Story 11 / Iceland: the smallest positive maximum

Observation: Temperature-change maximum 0.00006609375 from three observations, the smallest positive country maximum among 140 observed countries.

Interpretation: The precision belongs to a processed signal. It is not a temperature measurement accurate to that many decimal places. A small value provides a specific contrast within the selected view, leaving absolute climate, vulnerability, and future change as different questions.

Story 12 / Russia: emissions and geographic responsibility

Observation: Emissions-related maximum 1.00000 from 273 observations, highest among 146 observed countries.

Interpretation: The strong local signal makes emission-related pressure visible. It is not a ranking of national totals, per-person emissions, or exposure. The surrounding atmosphere opens questions about activities and shared consequences, while attribution to gases, sectors, or particular facilities needs a more specific source definition. [S10]

Story 13 / Iran and Iraq: similar appearance, different support

Observation: Iran's emissions maximum is 0.48880 from 22 observations, third among 146 countries. Iraq's is 0.41472 from five, fifth among the same comparison.

Interpretation: Both invite attention to activities and environmental pressure, but they have different observational support. Their marks cannot independently identify an industrial cause. Visual proximity in magnitude should not substitute for an explanation of the source or a national emissions inventory.

Story 14 / Guinea-Bissau: low production signal, open vulnerability

Observation: Emissions maximum 0.17041 from one observation, close to the observed minimum of 0.17040. Four country maxima are strictly lower. Temperature and emotional observations are absent in the selected views.

Interpretation: Low emission-related signal does not imply little environmental vulnerability. One attributed point is also not a national total. Different kinds of quietness on the globe can reflect low observed values or absent evidence.

Story 15 / Burundi: an economic endpoint

Observation: GDP per person is US\$216.23 in 2024, lowest among 190 available values. The yellow signal reaches 1.0.

Context: The World Bank describes constraints including low-productivity agriculture and landlocked geography. [C2]

Interpretation: These conditions give context for economic production without quantifying their separate causal contributions. Output per person does not measure household income, resource distribution, or the intensity of personal suffering.

Story 16 / Luxembourg: production and residence

Observation: GDP per person is US\$137,781.68, third-highest among 190 available values. The yellow signal is approximately 0.1025.

Context: Cross-border workers contribute to production but live outside the resident population denominator. [C4]

Interpretation: This accounting distinction explains why the figure is not a typical salary. A high national ratio provides one view of resources while leaving distribution, housing costs, and household security unresolved.

Story 17 / Monaco: zero yellow and an emotional word

Observation: GDP per person is US\$288,001.57, the maximum, producing zero yellow intensity. The displayed Hurt score is 0.1309 from 27 news and 1,034 expression records.

Interpretation: The economic zero is an observed endpoint, not missing data or absence of pain. Emotional language and economic output concern different constructs. Their coexistence clarifies why the artwork uses several layers without requiring them to agree.

Story 18 / India: the denominator changes the question

Observation: GDP per person is US\$2,591.99, forty-sixth-lowest among 190 available values; the yellow signal is approximately 0.6548.

Interpretation: A large total economy need not have high output per person. The selected indicator divides production by population, while leaving inequality and poverty prevalence unmeasured. Combined with India's physical concentration, it invites questions about resources and care without calculating a causal relationship between the layers.

Story 19 / Yemen: a missing economic observation

Observation: GDP is unavailable in the selected 2024 roster. The emotional view still contains the word Grief, with score 0.1886 from 27 news and 1,041 expression records.

Interpretation: Coverage belongs to each source. A blank economic entry and an observed emotional category can coexist. The absence supplies no numerical estimate of hardship; it also does not justify replacing the missing GDP with zero or treating the absence as prosperity.

Story 20 / Burundi: a strong anger score from 30 texts

Observation: Anger score 0.5202, highest among 192 emotional country results, based on 28 news and two expression records. The source-family means receive 75/25 weighting.

Interpretation: The category includes outrage and moral injury, not simply aggressiveness. The result concerns selected language. Thirty texts cannot establish national character or explain what caused the score. The source mixture is part of the finding, not background that can be omitted.

Story 21 / Palau: the visible winner is not the strongest control

Observation: One news article supplies the result. Uncertainty leads the displayed pain categories at 0.1557; anger is 0.0105, the lowest anger score. Out-of-scope is higher at 0.6492.

Interpretation: The display excludes control categories when choosing its word. A prominent word therefore does not certify strong distress evidence. Palau makes both tiny sample size and the display's choice of eligible categories visible as interpretive limits.

Story 22 / Timor-Leste: a strong score from eight records

Observation: Anger score 0.4738, second-highest among 192 emotional country results, based on five news and three expression records.

Interpretation: A strong score can arise from a very small selected collection. The 75/25 source-family weighting also differs from treating all eight records equally. The result can prompt questions about the language represented, but it cannot establish residents' typical feelings or the historical causes of a national pattern.

Story 23 / Sao Tome and Principe: environmental concern in language

Observation: Eco-anxiety score 0.2008, highest among 192 country results, based on 27 news records and no expression records. The available news family receives full weight.

Interpretation: The category concerns language of environmental fear or anxiety. It is neither a climate sensor nor a survey of residents. Considering it beside the environmental layer distinguishes material change from the ways concern is expressed and reported.

Story 24 / Oman: bodily hurt in two kinds of evidence

Observation: Body-pain language score 0.1580, highest among 192 emotional results, from 27 news and 968 expression records. The separate physical maximum is 0.04351 across 36 condition-points.

Interpretation: Shared vocabulary does not make the quantities interchangeable. One is a weighted mean of text scores; the other is a local health-related maximum. Neither their magnitudes nor their denominators provide a direct cross-layer comparison of pain intensity.

Story 25 / Comoros: visibility depends on the source

Observation: GDP per person is US\$1,857.87, with yellow signal approximately 0.7010. The selected emotional collection has zero news and zero expression records for Comoros.

Interpretation: The emotional result is missing, not a measured zero. An economic entry cannot fill that gap because it measures something else. The example brings attention to whose information reaches each layer, and why visibility should not be confused with existence.

Interpretation, uncertainty, and questions worth keeping open

More records and more information

Effective sample size means the amount of independent information available for a particular inference. It can be smaller than the record count when articles repeat an event, publishers share material, or authors contribute multiple posts. Exact duplicate removal helps with one form of repetition but does not remove thematic or organizational dependence. A million articles do not automatically supply a million independent observations of social experience.

No population confidence intervals are established by the country means in this guide. Treating all records as independent draws from residents would underestimate uncertainty and misidentify the target population. Statistical uncertainty around a textual mean also differs from uncertainty about whether the metric captures the intended human construct.

Association, causation, and the ecological leap

An **ecological inference** uses group-level observations to make a claim about individuals. The leap is unsafe when a country's average or maximum is assigned to every resident. A country with a strong health signal need not have uniformly high individual risk. A country with an anger label need not contain a high proportion of angry residents.

Co-location of two layers does not identify causation. Economic conditions, health, climate, and language may interact, but their source periods and geographic supports differ. Estimating a relationship would first require a specified question, compatible measurements, appropriate comparison groups, and attention to alternative explanations. The artwork invites those questions without already answering them statistically.

Five useful expert questions

- **What is the target of the claim?** A set of texts, a publisher system, a population, or a lived experience would each require a different evidential basis.
- **What changes under another aggregation?** A mean, maximum, source-balanced result, and record-weighted result emphasize different features of the same material.
- **Whose absence matters?** Missing countries, poorly covered languages, excluded publishers, and people who do not post leave different gaps.
- **How stable is a visible label?** A winning category can depend on source composition, permitted categories, time selection, and model calibration.
- **What would support a causal story?** Context can motivate a hypothesis, but explanatory attribution needs evidence beyond an evocative spatial coincidence.

An artistic framing can emphasize mutual dependence and care. A scientific framing can emphasize measurement and uncertainty. An everyday framing can begin with movement, home, work, or concern for another person. These are different ways to encounter the same evidence, with room for visitors to contribute meaning without being treated as data points.

Cross-disciplinary glossary

Term	Working meaning
Aggregation	Combining observations into a summary; its weighting defines the question answered.
Calibration	Agreement between predicted probabilities and reference frequencies in a specified population of cases.
Construct	The broader concept an indicator attempts to represent.
Country attribution	The rule associating a record with a country; news uses publisher origin.
Domain shift	A change between the kinds of material used for training, evaluation, and application.
Effective sample size	Independent information for an inference, which can be less than the number of records.
Embedding	A numerical representation of text learned by a language model.
Exposure	Contact of a person or system with an environmental agent, distinct from emitted quantity.
Gabriel graph	A proximity graph using an empty ball with each proposed edge as its diameter.
Macro-F1	Equal-weight average of category F1 scores, balancing precision and recall.
Measurement invariance	Comparable meaning and performance of a measure across groups.
Multi-label classification	Allowing several categories to apply to one input.
Proxy	A specified indicator representing part of a broader concept.
Scope gating	Weighting category outputs by whether the input is relevant to distress.
Solastalgia	Distress connected with unwanted environmental change to a familiar home.
Spatial support	The location or area to which an observation refers.
Synthetic data	Constructed examples with intended properties, distinct from naturally occurring testimony.
Threshold	A boundary turning a continuous score into a decision.

References and further reading

Public references identify source programmes and contextual concepts. Exact project selection, model, and exhibition counts were checked against the project's research records and selected exhibition inputs on 9 September 2026. A source institution's public overview does not, by itself, validate the project's particular subset or transformation. Unresolved release details remain identified above.

Public dataset sources

- [S1] **Stanford OVAL CCNews**. Dataset card, upstream scope and crawl/publication-date distinction. [Source page](#).
- [S2] **Twitter Sentiment by Country, wjia26**. Country-associated social-media material. [Source page](#).
- [S3] **DEMOTEC participatory-budgeting tweets**. Zenodo record, version 1, 4 February 2025; collection topic, languages, and 2015-2021 period. [Source record](#).
- [S4] **Yachay Seasons**. Coordinate-linked social-media dataset card and upstream scope. [Source page](#).
- [S5] **CFPB Consumer Complaint Database**. Institutional explanation of the complaint collection. [Source page](#).
- [S6] **CMU GeoText**. Geotagged microblog corpus; release 2010-10-12 and associated research. [Source page](#).
- [S7] **IHME Global Burden of Disease**. Health-estimation study. Background on the health source. [Source overview](#).
- [S8] **WorldPop Global 2000-2020**. Population product family; UN-adjusted approximately 1 km data. [Product overview](#). [Population data](#).
- [S9] **Berkeley Earth**. Temperature datasets and product descriptions. [Data overview](#).
- [S10] **Climate TRACE**. Emissions data and measurement definitions. [Data downloads](#). [Inventory definitions](#).
- [S11] **World Bank**. GDP per capita, current US dollars. [Indicator](#). The exhibition uses the 2024 observations.

Context for selected stories

- [C1] **IPCC AR6 Working Group I, Chapter 4**. Regional amplification and the interacting processes underlying Arctic warming. General regional context, not an explanation of every individual mark. [Chapter](#).
- [C2] **World Bank, Burundi overview**. Economic structure, landlocked geography, and development constraints. Accessed 9 September 2026; contextual overview, not a decomposition of the 2024 GDP figure. [Country overview](#).
- [C3] **WHO, Low back pain, 19 June 2023**. Age patterns and global burden; not a Japan-specific explanation of the exhibition's strongest local signal. [Fact sheet](#).
- [C4] **STATEC, Luxembourg in figures and analysis, Kaleidoscope 2006**. Explains the GDP-per-capita effect of cross-border workers. Used for the accounting principle, not current numerical estimates. [Publication](#).
- [C5] **WHO, Osteoarthritis, 14 July 2023**. Condition definition and general risk factors. [Fact sheet](#).
- [C6] **MedlinePlus, Rheumatoid arthritis**. Immune-related condition definition. [Medical encyclopedia](#).
- [C7] **WHO, Migraine and other headache disorders**. The everyday and public-health burden of headache conditions. [Fact sheet](#).
- [C8] **Statistics Bureau of Japan**. Population estimates for 1 October 2024, including the share aged 65 and over. [Population overview](#).
- [S12] **BAAI BGE-M3**. Background on the pretrained multilingual encoder used by the classifier. [Model overview](#).